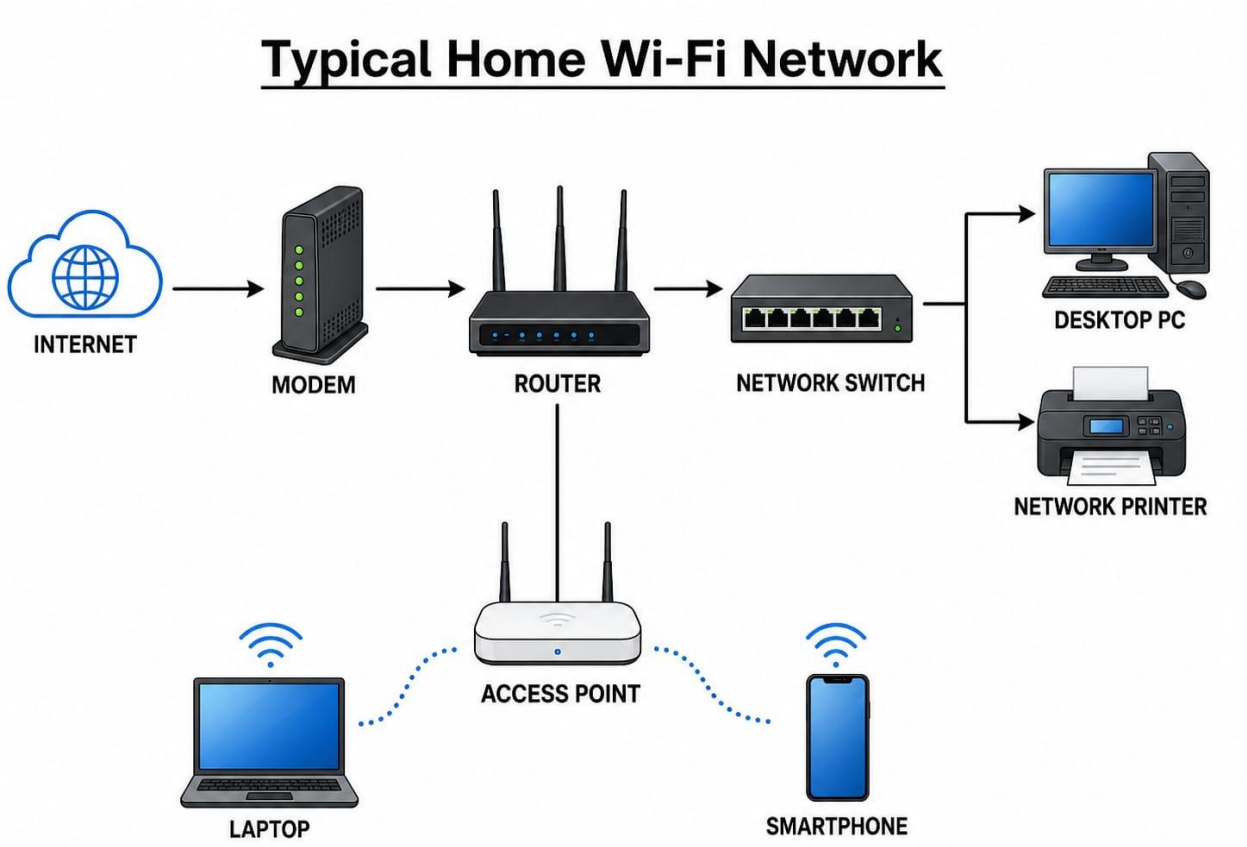
Q1. Draw a labeled diagram of a typical home Wi-Fi setup.



Q2. Main Functions of Networking Hardware Devices

1. Modem: A modem connects our home network to the internet. Example: A modem used in a home broadband connection.

2. Router: A router connects different networks and shares the internet with multiple devices. Example: A Wi-Fi router at home.

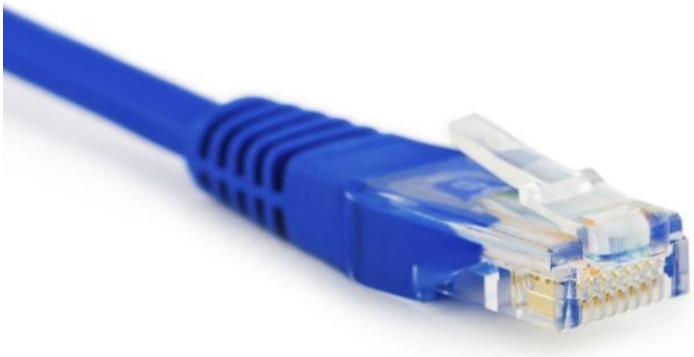
3. Switch: A switch connects multiple wired devices within a local area network (LAN). Example: Connecting computers in a computer lab.

4. Access Point: An access point provides wireless network connectivity to devices. Example: Providing Wi-Fi in a school or office.

5. Network Cable: A network cable carries data between network devices. Example: An Ethernet cable connecting a computer to a router.

Q3. Types of Network Cables

1. Ethernet Cable



It is used to connect computers, routers, and switches in a LAN.

2. Coaxial Cable



It is commonly used for cable TV and some internet connections.

3. Fiber Optic Cable



It transmits data using light and is used for high-speed internet and long-distance communication.

Q4. Difference Between Router and Switch

Point	Router	Switch

1. Main use	Connects different networks.	Connects devices within a LAN.
2. Data handling	Forwards packets between networks using IP addresses.	Forwards frames within a LAN using MAC addresses.
3. Typical location	Homes, offices, and internet gateways.	Computer labs, offices, and server rooms.
4. Internet connection	Commonly connects a local network to the internet.	Usually connects devices within the local network.
5. Example	Home Wi-Fi router.	Switch connecting computers in a lab.